resistor allows for board space savings as it can replace multiple standard parallel resistors in smaller case sizes. With a resistance range from 1Ω to 10MΩ, including a 0Ω jumper option, the resistor offers tolerance options of ±0.5%, ±1%, and ±5%, along with TCR values of ±100ppm/K and ±200ppm/K. It operates at a voltage of 150V and has a wide operating temperature range from -55°C to +155°C. The RCS0805 e3 is RoHS-compliant, halogen-free, and compatible with various assembly and soldering processes.

Vishay Intertechnology
https://www.vishay.com

EMBEDDED

5G NR+Wi-Fi 6E SoC

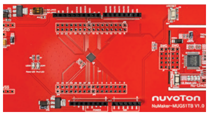
Qualcomm QCM4490 is an advanced 5G NR+Wi-Fi 6E SoC with Bluetooth 5.2 support. This versatile module offers reduced latency and increased responsiveness for voice and video applications, ensuring seamless connectivity in challenging environments.

With its utilisation of the 6GHz spectrum, it extends advanced Wi-Fi 6 features for reliable performance in congested areas. The QCM4490 incorporates on-device processing, supporting efficient data analysis between devices and the cloud. It boasts impressive speeds up to 3.6Gbps, powered by Qualcomm's 4K QAM and 160MHz channel support. Additionally, it features an audio AI accelerator for enhanced voice quality, an advanced camera ISP for superior image capture, and Gen 9 V5 GNSS supporting various satellite systems. The QCM4490 presents an ideal solution for industrial handheld devices, retail POS systems, industrial computing devices, control and automation, and industrial and personal security panels.

Qualcomm
https://www.qualcomm.com

Flash embedded MCU

The low power MUG51 series from Nuvoton Technology is an efficient flash embedded microcontroller solution for battery-free devices. These microcontrollers feature low current

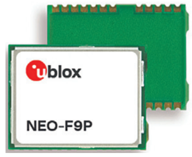


consumption, drawing just 200µA during CPU power on before flash memory initialisation, making them ideal for devices powered by magnetic fields such as stylus pens and RFID cards. With an extensive range of peripherals including UARTs, SPI interfaces, I²C interfaces, and analogue comparators, users can achieve seamless connectivity and enhanced functionality. The series incorporates power management functions and supports wake-up from power down mode through various triggers, thus, ensuring efficient operation while conserving power.

Nuvoton Technology
https://www.nuvoton.com

High-precision GNSS module

u-blox has introduced two new high-precision GNSS positioning modules, the NEO-F9P and ZED-F9P-15B, based on the F9 platform, catering to precise navigation and automation in various industries. The NEO-F9P module is designed for applications requiring accurate positioning and robust performance, supporting concurrent reception of multiple satellite systems



like GPS, Galileo, and BeiDou. It incorporates multi-band L1/L5 RTK technology, ensuring centimetre-level accuracy in seconds, crucial for industries like smart antennas, UAVs, and mobile robotics. The NEO-F9P module stands out for its compact form factor, being 50% smaller than the regular u-blox ZED form factor, making it highly versatile for space-constrained applications. Moreover, it boasts low power consumption, making it suitable for battery-powered devices. The module is also compatible with the ANN-MB1 antenna, further enhancing its usability. Additionally, the ZED-F9P-15B GNSS module targets the mobile robotics market, offering an L1/L5 alternative to L1/L2 bands, ensuring increased positioning accuracy and reliability, especially in challenging environments.

u-Blox
https://www.u-blox.com

Positioning solution employing computer vision

Point One Navigation and Septentrio have collaborated to launch Polaris, a cutting-edge correction network that

enables precise GPS and computer vision-based localisa-